

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-008 (W008) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W008
Well Name	SYN-Assam-Duliajan-008
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.307921 N / 95.286728 E
Rig	SYN-RIG-N1600-02
Spud Date	24-Mar-2018
Final Report Date	05-Apr-2018
Daily Reports in Set	13 (Day 01 to Day 13)
Total Depth Reached	3,690 m
Primary Formation	Sandstone
Events Reported in Set	8

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	24-Mar-2018	0	545	545	Claystone	Wellbore Instability, Cementing Issue
02	25-Mar-2018	545	865	320	Claystone / Shale	—
03	26-Mar-2018	865	1,295	430	Shale	Cementing Issue
04	27-Mar-2018	1,295	1,595	300	Shale	—
05	28-Mar-2018	1,595	1,825	230	Shale	Wellbore Instability
06	29-Mar-2018	1,825	2,145	320	Shale / Sandstone	—
07	30-Mar-2018	2,145	2,445	300	Sandstone	—
08	31-Mar-2018	2,445	2,720	275	Sandstone	Mud Loss
09	01-Apr-2018	2,720	2,890	170	Sandstone	Mud Loss
10	02-Apr-2018	2,890	3,130	240	Sandstone	—
11	03-Apr-2018	3,130	3,360	230	Sandstone / Limestone	—
12	04-Apr-2018	3,360	3,575	215	Limestone	Wellbore Instability
13	05-Apr-2018	3,575	3,690	115	Limestone	Casing Issue

DAILY DRILLING REPORT — DAY 01						W008 · 24-Mar-2018																					
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-03-24 Day Number: 1 Start Depth: 0 m End Depth: 545 m Drilled Interval: 545 m Formation: Claystone Mud Weight: 9.40 ppg ROP: 32.3 m/hr RPM: 143 WOB: 7.1 klbf Events: Wellbore Instability at 412 m (High); Cementing Issue at 524 m (High) NPT: 44.3 hours																											
Operations performed. - Rig up completed and well spudded at 24-Mar-2018. Drilled 17-1/2" hole from surface to 545 m. - Deviation survey taken; hole angle within tolerance. - Wellbore Instability handled at 412 m — 25.4 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval. - Cementing Issue handled at 524 m — 18.9 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.																											
Geological observations. Cuttings throughout the interval are claystone, consistent with the 0–838 m unit logged in the offset wells of this cluster.																											
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 9.40 ppg, properties held within programme; no losses or gains recorded.																											
Equipment / operational observations. Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.																											
Events reported this day. <table><tr><th>Event ID</th><th>Depth (m)</th><th>Formation</th><th>Type</th><th>Severity</th><th>Hr</th><th>Operational Impact</th></tr><tr><td>E0043</td><td>412</td><td>Claystone</td><td>Wellbore Instability</td><td>High</td><td>25.4</td><td>Significant non-productive time; remedial operations and additional materials required.</td></tr><tr><td>E0044</td><td>524</td><td>Claystone</td><td>Cementing Issue</td><td>High</td><td>18.9</td><td>Significant non-productive time; remedial operations and additional materials required.</td></tr></table>							Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact	E0043	412	Claystone	Wellbore Instability	High	25.4	Significant non-productive time; remedial operations and additional materials required.	E0044	524	Claystone	Cementing Issue	High	18.9	Significant non-productive time; remedial operations and additional materials required.
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact																					
E0043	412	Claystone	Wellbore Instability	High	25.4	Significant non-productive time; remedial operations and additional materials required.																					
E0044	524	Claystone	Cementing Issue	High	18.9	Significant non-productive time; remedial operations and additional materials required.																					
End-of-day status. Depth 545 m in Claystone. Drilling ahead. Non-productive time this tour 44.3 hr.																											

DAILY DRILLING REPORT — DAY 02						W008 · 25-Mar-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-03-25 Day Number: 2 Start Depth: 545 m End Depth: 865 m Drilled Interval: 320 m Formation: Claystone / Shale Mud Weight: 10.00 ppg ROP: 29.8 m/hr RPM: 136 WOB: 10.6 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 320 m of section from 545 m to 865 m (16.5 hr on bottom). - Slipped and cut drilling line; routine BOP function test carried out. - Pulled out of hole, ran 13-3/8" casing to 840 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.						
Geological observations. Formation change crossed during the tour: Claystone to 838 m, Shale to 2,020 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 10.00 ppg, properties held within programme; no losses or gains recorded.						

DAILY DRILLING REPORT — DAY 03

W008 · 26-Mar-2018

Well ID: W008
Well Name: SYN-Assam-Duliajan-008
Report Date: 2018-03-26
Day Number: 3
Start Depth: 865 m
End Depth: 1295 m
Drilled Interval: 430 m
Formation: Shale
Mud Weight: 10.62 ppg
ROP: 16.7 m/hr
RPM: 129
WOB: 15.5 klbf
Events: Cementing Issue at 1180 m (Medium)
NPT: 10.2 hours

Operations performed.

- Drilled 430 m of Shale from 865 m to 1,295 m (12.3 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.
- Cementing Issue handled at 1,180 m — 10.2 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 838–2,020 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 10.62 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0046	1,180	Shale	Cementing Issue	Medium	10.2	Partial non-productive time; drilling suspended while the well was treated and conditioned.

End-of-day status.

Depth 1,295 m in Shale. Drilling ahead. Non-productive time this tour 10.2 hr.

DAILY DRILLING REPORT — DAY 04

W008 · 27-Mar-2018

Well ID: W008
Well Name: SYN-Assam-Duliajan-008
Report Date: 2018-03-27
Day Number: 4
Start Depth: 1295 m
End Depth: 1595 m
Drilled Interval: 300 m
Formation: Shale
Mud Weight: 11.14 ppg
ROP: 16.1 m/hr
RPM: 126
WOB: 17.6 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 300 m of Shale from 1,295 m to 1,595 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 838–2,020 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.14 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,595 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 05						W008 · 28-Mar-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-03-28 Day Number: 5 Start Depth: 1595 m End Depth: 1825 m Drilled Interval: 230 m Formation: Shale Mud Weight: 11.56 ppg ROP: 15.1 m/hr RPM: 121 WOB: 19.7 klbf Events: Wellbore Instability at 1673 m (High) NPT: 23.9 hours						
Operations performed. - Drilled 230 m of Shale from 1,595 m to 1,825 m (2.0 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Wellbore Instability handled at 1,673 m — 23.9 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.						
Geological observations. Cuttings throughout the interval are shale, consistent with the 838–2,020 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.56 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. BOP drill carried out with the crew; response time satisfactory.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0045	1,673	Shale	Wellbore Instability	High	23.9	Significant non-productive time; remedial operations and additional materials required.
End-of-day status. Depth 1,825 m in Shale. Drilling ahead. Non-productive time this tour 23.9 hr.						

DAILY DRILLING REPORT — DAY 06						W008 · 29-Mar-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-03-29 Day Number: 6 Start Depth: 1825 m End Depth: 2145 m Drilled Interval: 320 m Formation: Shale / Sandstone Mud Weight: 11.83 ppg ROP: 15.5 m/hr RPM: 117 WOB: 22.1 klbf Events: None NPT: 0.0 hours						
Operations performed. - Drilled 320 m of section from 1,825 m to 2,145 m (16.5 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Pulled out of hole, ran 9-5/8" casing to 2,020 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 720 m). Waited on cement and pressure tested.						
Geological observations. Formation change crossed during the tour: Shale to 2,020 m, Sandstone to 3,271 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 11.83 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Mud pump No. 2 packing changed during routine circulation; no lost time.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status.						

DAILY DRILLING REPORT — DAY 07	W008 · 30-Mar-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-03-30 Day Number: 7 Start Depth: 2145 m End Depth: 2445 m Drilled Interval: 300 m Formation: Sandstone Mud Weight: 12.14 ppg ROP: 16.3 m/hr RPM: 115 WOB: 23.3 klbf Events: None NPT: 0.0 hours	
Operations performed. - Drilled 300 m of Sandstone from 2,145 m to 2,445 m (22.5 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored.	
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,020–3,271 m unit logged in the offset wells of this cluster.	
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 12.14 ppg, properties held within programme; no losses or gains recorded.	
Equipment / operational observations. BOP drill carried out with the crew; response time satisfactory.	
Events reported this day. None. Operations proceeded within the planned parameter envelope.	
End-of-day status. Depth 2,445 m in Sandstone. Drilling ahead. No non-productive time this tour.	

DAILY DRILLING REPORT — DAY 08

W008 · 31-Mar-2018

Well ID: W008
Well Name: SYN-Assam-Duliajan-008
Report Date: 2018-03-31
Day Number: 8
Start Depth: 2445 m
End Depth: 2720 m
Drilled Interval: 275 m
Formation: Sandstone
Mud Weight: 12.54 ppg
ROP: 15.6 m/hr
RPM: 112
WOB: 24.8 klbf
Events: Mud Loss at 2669 m (High)
NPT: 11.1 hours

Operations performed.

• Drilled 275 m of Sandstone from 2,445 m to 2,720 m (11.4 hr on bottom).

• Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

• Mud Loss handled at 2,669 m — 11.1 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 2,020–3,271 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.54 ppg after treatment.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0047	2,669	Sandstone	Mud Loss	High	11.1	Significant non-productive time; remedial operations and additional materials required.

End-of-day status

Prototyping Training Record - Synthetic demonstration data

Depth 2,720 m in Sandstone. Drilling ahead. Non-productive time this tour 11.1 hr.

Page 5 of 8

DAILY DRILLING REPORT — DAY 09						W008 · 01-Apr-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-04-01 Day Number: 9 Start Depth: 2720 m End Depth: 2890 m Drilled Interval: 170 m Formation: Sandstone Mud Weight: 12.84 ppg ROP: 15.0 m/hr RPM: 108 WOB: 26.8 klb Events: Mud Loss at 2735 m (High) NPT: 16.2 hours						
Operations performed. - Drilled 170 m of Sandstone from 2,720 m to 2,890 m (6.3 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Mud Loss handled at 2,735 m — 16.2 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,020–3,271 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.84 ppg after treatment.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0049	2,735	Sandstone	Mud Loss	High	16.2	Significant non-productive time; remedial operations and additional materials required.
End-of-day status. Depth 2,890 m in Sandstone. Drilling ahead. Non-productive time this tour 16.2 hr.						

DAILY DRILLING REPORT — DAY 10						W008 · 02-Apr-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-04-02 Day Number: 10 Start Depth: 2890 m End Depth: 3130 m Drilled Interval: 240 m Formation: Sandstone Mud Weight: 13.18 ppg ROP: 13.7 m/hr RPM: 106 WOB: 28.6 klb Events: None NPT: 0.0 hours						
Operations performed. - Drilled 240 m of Sandstone from 2,890 m to 3,130 m (22.5 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 2,020–3,271 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.18 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. BOP drill carried out with the crew; response time satisfactory.						
Events reported this day. None. Operations proceeded within the planned parameter envelope.						
End-of-day status. Depth 3,130 m in Sandstone. Drilling ahead. No non-productive time this tour.						

DAILY DRILLING REPORT — DAY 11	W008 · 03-Apr-2018
Well ID: W008 Well Name: SYN-Assam-Duliajan-008 Report Date: 2018-04-03 Day Number: 11 Start Depth: 3130 m End Depth: 3360 m Drilled Interval: 230 m Formation: Sandstone / Limestone Mud Weight: 13.68 ppg ROP: 11.2 m/hr RPM: 99 WOB: 32.7 klbf Events: None NPT: 0.0 hours	
Operations performed. - Drilled 230 m of section from 3,130 m to 3,360 m (22.5 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored.	
Geological observations. Formation change crossed during the tour: Sandstone to 3,271 m, Limestone to 3,690 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.	
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.68 ppg, properties held within programme; no losses or gains recorded.	
Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime.	
Events reported this day. None. Operations proceeded within the planned parameter envelope.	
End-of-day status. Depth 3,360 m in Sandstone. Drilling ahead. No non-productive time this tour.	

DAILY DRILLING REPORT — DAY 12

W008 · 04-Apr-2018

Well ID: W008
Well Name: SYN-Assam-Duliajan-008
Report Date: 2018-04-04
Day Number: 12
Start Depth: 3360 m
End Depth: 3575 m
Drilled Interval: 215 m
Formation: Limestone
Mud Weight: 14.19 ppg
ROP: 7.1 m/hr
RPM: 89
WOB: 36.2 klbf
Events: Wellbore Instability at 3513 m (Critical)
NPT: 36.8 hours

Operations performed.

- Drilled 215 m of Limestone from 3,360 m to 3,575 m (2.0 hr on bottom).
- Deviation survey taken; hole angle within tolerance.
- Pulled out of hole, ran 7" casing to 3,450 m and cemented as programmed (Tail 15.8 ppg, TOC 1,870 m). Waited on cement and pressure tested.
- Wellbore Instability handled at 3,513 m — 36.8 hr non-productive time. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,271–3,690 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 14.19 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0048	3,513	Limestone	Wellbore Instability	Critical	36.8	Major non-productive time; section objectives affected and specialist intervention required.

DAILY DRILLING REPORT — DAY 13	W008 · 05-Apr-2018
--------------------------------	--------------------

Well ID: W008
Well Name: SYN-Assam-Duliajan-008
Report Date: 2018-04-05
Day Number: 13
Start Depth: 3575 m
End Depth: 3690 m
Drilled Interval: 115 m
Formation: Limestone
Mud Weight: 14.43 ppg
ROP: 6.7 m/hr
RPM: 86
WOB: 36.9 klbf
Events: Casing Issue at 3637 m (Low)
NPT: 5.6 hours

Operations performed.

- Drilled 115 m of Limestone from 3,575 m to 3,690 m (10.9 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.
- Pulled out of hole, ran 5" liner casing to 3,690 m and cemented as programmed (Tail 15.8 ppg, liner lap tested). Waited on cement and pressure tested.
- Reached driller's total depth of 3,690 m. Circulated hole clean, logged and secured the well.
- Casing Issue handled at 3,637 m — 5.6 hr non-productive time. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,271–3,690 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 14.43 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0050	3,637	Limestone	Casing Issue	Low	5.6	Minor non-productive time; operations resumed after routine parameter adjustment.

End-of-day status.

Depth 3,690 m in Limestone. 5" liner casing cemented at 3,690 m; waiting on cement.